

A Multidisciplinary Collaboration to Reduce Vaping in Adolescents

Elisha Tribble and Daphnee Stewart

E-cigarette use, known as vaping, among adolescents is becoming a nationwide concern. According to the Centers for Disease Control and Prevention (CDC), in 2020, “19.6% of high school students (3.02 million) and 4.7% of middle school students (550,000) reported current e-cigarette use” (Wang et al., 2020, p. 1). Tobacco use has been identified as an objective from the U.S. Department of Health and Human Services Healthy People 2030. The Healthy People 2030 goal related to Tobacco Use is to “Reduce illness, disability, and death related to tobacco use and secondhand smoke” with a specific objective to “Eliminate the initiation of the use of cigarettes among adolescents and young adults” (Healthy People 2030, 2022). As students enter their adolescent years, they are more likely to try and continue the use of e-cigarettes and vaping products because they are easily concealed and do not have the typical tobacco odor one associates with cigarette use (Chadi et al., 2019). Approximately one-third of high school students also view e-cigarettes as less harmful and less addictive than traditional cigarettes, and are unaware that e-cigarettes not only contain nicotine, but can be mixed with marijuana, tetrahydrocannabinol (THC), and hash oil (Chung & Bayakly, 2018).

The negative effects of e-cigarette use are numerous. E-cigarette products contain elevated benzoic acid

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Background: E-cigarette use among adolescents is a growing concern. Approximately 20% of high school students and 5% of middle school students across the United States report the use of e-cigarettes (Wang et al., 2020). To address this concern, the authors developed a curriculum about vaping and collaborated with the Georgia Sheriffs' Association to implement it into the existing Choosing Healthy Activities and Methods Promoting Safety (CHAMPS) program regarding the dangers of e-cigarette use and vaping to help decrease the usage of e-cigarettes in adolescents. The purpose of this project was to develop, implement, and evaluate the new vaping curriculum incorporated in the CHAMPS program.

Methods: The curriculum was created to be cohesive with the current CHAMPS curriculum in use across Georgia. The new vaping education was integrated into the existing program and introduced to 64 Georgia Sheriffs' Association deputies to educate and prepare them to use the new education in the classroom. Following the intervention, 54 deputies completed an anonymous emailed survey to determine their perceptions of benefit, impact, and sustainability of the new curriculum.

Findings: Results of the survey showed that the deputies perceived the new vaping curriculum to be beneficial, has an impact on the adolescent population, and is sustainable for continued use.

Discussion: Through a collaborative effort with the Georgia Sheriffs' Association, this project has broadened the current CHAMPS curriculum, and educated younger adolescents on the dangers of vaping and e-cigarette use. The anticipated impact of this early educational intervention is to see an overall decrease in e-cigarette usage in adolescents. Health care providers can partner and collaborate with community or state agencies to have an impact on population health or reach a target audience.

Keywords: Adolescents, vaping, e-cigarettes, substance use.

and nicotine salt levels, producing high nicotine concentrations in the vaping solution. Exposure to these high doses of nicotine contributes to adolescents becoming nicotine-dependent and addicted at an early age (Patterson et al., 2020). During the adolescent years, as the brain is undergoing development, nicotine use can affect the neurologic pathways that control attention and learning, can contribute to mood disorders, and may cause impulse control issues in this age group (Patterson et al., 2020). Vaping increases the possibility of other risky behaviors, such as illicit drug

use, alcohol use, and cigarette smoking (Chadi et al., 2019; Hamberger & Halpern-Felsher, 2020; Patterson et al., 2020). Physical effects are also present because there is increasing evidence that vaping aerosolized liquids have a detrimental effect on the lungs. E-cigarette-associated lung injury (EVALI), a life-threatening, acute respiratory illness, along with chronic respiratory conditions, such as decreased mucociliary clearance, impaired pulmonary function, and alveoli changes, are long-term harmful effects resulting from e-cigarette use (Huey et al., 2020).

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Therefore, it is important to address the use of e-cigarettes and vaping at an early age to deter the use of these products in the adolescent years.

Purpose

E-cigarette use is specifically a current public health concern in Georgia, with one in four high school students having tried an e-cigarette (Chung & Bayakly, 2018). Furthermore, data show the increasing trend of e-cigarette usage each year in high school, with 9th grade students averaging 20% usage and 12th grade students averaging 32.2% usage (Chung & Bayakly, 2018). To address the usage among students in Georgia, a collaboration was established with the Georgia Sheriffs' Association, which currently delivers the Choosing Healthy Activities and Methods Promoting Safety (CHAMPS) program. This established program is delivered each school year to 5th and 6th grade students in over 90 counties in the state to educate students on the dangers of tobacco, drug, and alcohol use. Prior to this project, there was no information related to vaping in the existing program. The purpose of this project was to develop, implement, and evaluate the new vaping curriculum incorporated in the CHAMPS program. By creating and implementing an education intervention in the existing program, the goal was to establish early intervention, decrease the risk of vaping in adolescents, and achieve the Healthy People 2030 goal of eliminating the initiation of e-cigarette use in the adolescent population.

Review of the Literature

Search Methodology

This literature review included synthesized findings from systematic reviews, meta-analyses, and cross-sectional studies. An integrative review of the literature was conducted using the electronic databases of CINAHL Plus with full text, Medline, and Google Scholar. Key search terms used were *adolescent and vaping education, tobacco and vaping cessation education, prevention*

of substance abuse, educational interventions and vaping, e-cigarette education and interventions, and vaping/e-cigarette/substance use.

Inclusion Criteria

Inclusion criteria were articles that discussed educational interventions to reduce substance use and e-cigarette use in adolescents or youth. Studies were included in the review if articles were 1) published in peer-reviewed journals, 2) written in the English language, 3) published in the United States and/or internationally, and 4) published from 2010 to 2021.

Exclusion Criteria

Exclusion criteria included articles that did not discuss vaping, e-cigarette, or substance use, or target educational interventions for youth and adolescents. Articles were excluded if they did not mention adolescents or youth because the project focused on a specific population. Further, articles that did not discuss substance use, e-cigarettes, or vaping intervention were excluded because the purpose of this study was to find proven, effective interventions to help prevent and reduce adolescent vaping.

Synthesis of Findings

Ten articles were identified as meeting criteria for the review. Four articles were systematic review research design, and six articles were cross-sectional research design. After reviewing the literature, themes identified that were effective in reducing the incidence of vaping in adolescents and youth are school-based education, positive youth development (PYD) and self-regulation, and collaboration with school clinics and community resources.

School-Based Education

Education is an effective way to combat the vaping epidemic among adolescents. Vaping and smoking prevention can occur when school-based education starts in upper elementary or middle school and continues through high school (Flay, 2009; Gaiha et al., 2021; Kelder et al., 2020). When providing education through a school-based cur-

riculum, teachers and administration deliver the education and must know the dangers of vaping and e-cigarette use and tobacco products. This knowledge allows educators to help deter adolescents from vaping and smoking (Lazaro et al., 2021). Some schools receive funding and offer tobacco use prevention education (TUPE) to help deliver tobacco prevention education. Schools with TUPE funding are more likely to have lower tobacco use among adolescents (McMenamin et al., 2018).

Positive Youth Development and Self-regulation

Positive youth development and self-regulation are other interventions that can reduce substance abuse. School-based curriculum alone can be ineffective for the long-term prevention of smoking in adolescents and reducing the number of adolescents who currently smoke or vape if not combined with other effective measures (Backinger et al., 2003; Chaplin et al., 2020). When a school-based curriculum is combined with PYD, it provides youth with the foundation to build effective relationships and skills to develop positive experiences and positive relationships that will enable auto self-regulation and positive assets. Self-regulation and positive assets help create an environment that enables youth to avoid substance abuse. These skills allow adolescents to learn and build ways to manage stress, which can reduce their risk of vaping (Bonell et al., 2015).

Collaboration with School Clinics and Community Resources

Collaboration with school-based health clinics in conjunction with school-based curriculum is another effective measure to reduce vaping. The primary care provider in the school health clinic can offer behavioral interventions that will improve adolescents and youth outcomes regarding vaping and smoking initiation and cessation (Lazaro et al., 2021; Peirson et al., 2016). The provider can play a significant role in helping to reduce vaping in adolescents by educating and preparing adolescents to anticipate

situations where vaping may arise, such as in social settings, through peer pressure, and at school. The provider can also help encourage PYD and self-regulation. Preparing students by talking through scenarios and how to respond can help the student reject the offer of someone asking them to vape. Second, the provider needs to ask the student about vaping and substance abuse. Asking direct questions along with when, where, and why questions will allow the provider the opportunity to educate and guide the adolescent on any risky behavior. The provider can address factors that make the education successful: peer pressure, feelings of invincibility, and limited ability to think abstractly (Patterson et al., 2020). Furthermore, working with community partners, such as existing state programs that educate teens and adolescents on the dangers of smoking and drinking, allows for easy incorporation of vaping education to review the dangers of vaping (Gaiha et al., 2021; Patterson et al., 2020). By working with programs already in place and collaborating with other groups, a school health clinic can share health care knowledge and risks of vaping to achieve the Healthy People 2030 goal to reduce vaping in adolescents.

School-based vaping educational programs are effective in reducing the incidence of vaping and e-cigarette use in adolescents. Furthermore, when education is paired with PYD and self-regulation, the risk of experimenting with vaping or e-cigarette use declines (Backinger et al., 2003; Chaplin et al., 2020). These mechanisms allow youth to develop coping skills to create an environment that will help them avoid substance use. Finally, by collaborating with school-based health clinics and educators, interventions and education can continue in the class and school health clinic. With the combination of health care providers, education, self-regulation, positive social relationships, and support systems, we believe the incidence and prevalence of vaping and e-cigarette use can be reduced in adolescents.

When performing the literature review, a gap in the literature was noted. There are no programs specific to 5th and 6th grade students regarding vaping education and interventions to reduce vaping. Practice implications based on the review of the literature indicate educational intervention, positive youth development, and collaboration were effective measures to reduce the incidence of vaping in youth. School-based programs can help combat the vaping epidemic in young individuals and promote an environment that teaches youth how to decrease stress and reduce peer pressure. Collaboration with educators and school-based health clinics can be an additional way to reduce vaping incidence through continual monitoring and educational interventions. Georgia currently offers a CHAMPS program that goes into schools to educate 5th and 6th grade students on tobacco, drugs, and alcohol. Incorporating a vaping curriculum into an already proven curriculum that helps deter young individuals from harmful behaviors was a way to begin an early intervention program. Thus, by broadening the CHAMPS curriculum and providing early intervention to adolescents, this project aimed to decrease the use of e-cigarettes and tobacco in adolescents.

Conceptual Framework

The theoretical framework chosen for this project was Pender's Health Promotion Model. Pender's Health Promotion Model can be used to improve an individual's well-being (Pender, 1984). Pender's multi-dimensional model focuses on three areas: 1) individual characteristics and experiences; 2) behavior-specific cognitions, and affect; and 3) behavioral outcomes (Pender, 1984). Pender's Health Promotion Model also describes how individuals interact within their environment as they pursue a healthy lifestyle. Every individual has unique personal characteristics and experiences that can affect their actions and lifestyle. Variables can impact behavior-specific cognitions and affect, and have motivational

significance on an individual. Nursing actions can modify the variable, where health-promoting behavior is the desired outcome, the endpoint of Pender's Health Promotion Model. When nursing actions modify the variables, a behavior change should improve an individual's health and quality of life (Pender, 1984).

This framework was chosen because e-cigarette and tobacco use can contribute to an individual's health. An individual's quality of life can be improved by preventing acute or chronic health conditions before they occur. E-cigarette and tobacco use can contribute to chronic lung disease and cancer. Health promotion and preventative care can provide individuals with improved autonomy in making better decisions for their health and improving their self-care. By introducing vaping education in early adolescence, the goal is to provide education to deter adolescents from this harmful behavior that could affect their health. Furthermore, using Pender's Health Promotion Model and understanding how interpersonal relationships play a role in behavior helped when creating the vaping education program. Understanding how adolescents interact with their peers and the pressures they face would allow a vaping educational program to inform adolescents of risks associated with e-cigarette use and how to withstand peer pressure.

Methods

Design

The project was an evidence-based program development and evaluation project. A curriculum related to vaping and e-cigarette education was developed to add to the existing CHAMPS curriculum. Following the program's implementation, a survey was administered to evaluate if participants perceived the curriculum was beneficial, had a perceived impact, and was sustainable. The survey also elicited feedback for improvement in future programming.

Participants

Participants included all Georgia Sheriffs' Association deputies who teach the CHAMPS curriculum in Georgia. Recruitment of study participants was coordinated with the director of the CHAMPS program. Participants were informed about the project by the director and were notified of the project at an annual conference when the new vaping curriculum was introduced. There were no gender, age, race, or religious exclusion criteria for participation in the project. At the end of the project, the program director emailed a survey to all program participants. The total number of eligible participants was 64 deputies; of those, 54 responded to the emailed survey.

Setting

This project took place in elementary and middle schools located in over 90 counties across Georgia, where the CHAMPS curriculum is currently active. The only resources needed for this project were the deputies who taught the vaping curriculum, and a vaping PowerPoint® and lesson plan created for this project. The CHAMPS curriculum did not have any educational information regarding vaping to help educate students on the dangers of e-cigarettes. Thus, stakeholders in this project were the Georgia Sheriffs' Association deputies who taught in the CHAMPS program, and their recognition and support for the need for additional curriculum regarding the dangers of vaping and e-cigarette use.

Intervention

The CHAMPS program delivers educational content to multiple elementary schools in 90 counties in the state of Georgia. Deputies from the CHAMPS program spend one hour each week for 12 weeks in each school one time a year to present information related to drugs, alcohol, bullying, ATV safety, gangs, graduation, and other information to help adolescents choose healthy behaviors that promote their safety. Due to time limitations in each school, the curriculum was developed to meet the needs of the CHAMPS program. The authors created the new vaping curriculum,

including a Microsoft™ PowerPoint® and a lesson plan, using several evidence-based materials to include information related to facts about e-cigarettes and vaping, dangers related to vaping, the risk of addiction, and marketing targeting adolescents (Addiction Policy Forum, 2016; Benowitz, 2010; Chadi et al., 2019; Chung & Bayakly, 2018; David Geffen School of Medicine at UCLA, 2017; Hamberger & Halpern-Felsher, 2020; Huey et al., 2020; Llamas, 2020; Nickels et al., 2014; Patterson et al., 2020; Penn Medicine, 2020; Stanford Medicine, n.d.; Truth Initiative, 2018; Wang et al., 2020). This curriculum was integrated into the existing CHAMPS curriculum and presented by a deputy using lesson plans and PowerPoint® presentations, with time for classroom discussion.

This new vaping curriculum was introduced to deputies in the CHAMPS program at an annual conference to educate presenters on the material to prepare them to use the new education in the classroom. Over the course of the following school year, the vaping curriculum was incorporated and presented along with the existing CHAMPS curriculum across the state of Georgia. At the end of the year, a survey was administered to deputies who presented the information to evaluate the program.

Data Collection

To measure outcomes of this project, an anonymous survey was developed and administered to evaluate the vaping and e-cigarette program. This survey was created by the author and administered through SurveyMonkey™. The survey consisted of three Likert-scale quantitative questions and three qualitative questions. The three quantitative questions were answered on a Likert scale with options of Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree. The questions were as follows: 1) The vaping educational content was beneficial to the adolescent population; 2) The vaping educational program enhanced adolescent learning on the dangers of vaping; and 3) The vaping educational program is sustainable for

continued future use. Participants were also asked three qualitative, open-ended questions: 1) What were the strengths of the vaping program?, 2) What were the weaknesses of the vaping program?, and 3) What would you change about the program? Participation took approximately 10 minutes to complete the survey. The proposed time allotted for the return of the completed survey was two weeks from the date of email dissemination.

Ethical Considerations

Institutional Review Board approval was obtained from Mercer University before the project began. A letter of support from the Georgia Sheriff's Office was also obtained prior to project initiation. Informed consent from participants was collected prior to completion of the survey. The survey did not contain identifying information, and all response data were confidential. Any data collected via survey were stored on a password-protected device.

Findings

Data from the anonymous survey were statistically analyzed using Spearman's Rank Correlation Coefficient (Rs) analysis, and a descriptive statistical analysis of the qualitative data was performed. To investigate the reliability of the Likert-scale items, an Rs was computed as a statistical measure of the strength of the relationship between the three Likert-scale items on the survey. An Rs value between 0.70 to 0.89 would show a strong correlation, and 0.90 to 1.00 would show an extraordinarily strong correlation between the three items. The Rs value for the three Likert-scale items on the survey were 0.8, 0.8, and 1.0, thus showing a strong correlation and that the items are internally consistent. Of the 84% of deputies completing the survey, the majority perceived the curriculum to be beneficial, enhanced adolescent learning, and sustainable for future use (see Table 1).

To investigate the validity of the qualitative, constructed-response (CR) questions, the descriptive cod-

Table 1.
Likert Scale Items (Reported as Percentages)

Question	Strongly Disagree	Disagree	Undecided	Agree	Strongly Agree
The vaping educational content was beneficial to the adolescent population.	0.00	3.70	11.11	40.74	44.44
The vaping educational program enhanced adolescent learning on the dangers of vaping.	0.00	3.70	12.96	42.59	40.74
The vaping educational program is sustainable for continued future use.	0.00	3.70	16.67	37.04	42.59
<i>Mean</i>	0.00	3.70	13.58	40.12	42.59

Table 2.
Descriptive Coding of Qualitative Data

Vaping Education	Example Quote
What were the strengths of the vaping program? <i>Informative</i>	“Explained and documented the hazards of vaping as compared to tobacco usage. Many of the target audience were completely unaware of these facts.” “Informing the kids that vaping is harmful. So many of them didn’t realize there’s nicotine in the vape and that it isn’t just water vapor!”
What were the weaknesses of the vaping program? <i>More Information Needed</i>	“Small amount of marketing examples.” “Would like to have videos for the kids.” “More visual illustrations.”
What would you change about the program? <i>Needed Changes/Edits</i>	“The information appeared to be way too advanced for the 5th grade population.” “Update information as it becomes known.” “I would add more graphics. Show the kids what vaping can do.”

ing method from Miles and colleagues (2020) was used. The descriptive coding assigned themes or labels that summarize a word or phrase. After performing descriptive coding of all responses to the CR questions, a list of categories was created with passages that fit each list of categories extracted and reviewed to create a narrative that describes that category or theme. See Table 2 for the qualitative data analysis and themes with some passages from responses received from the surveys.

Discussion

Data analysis shows participants perceived the addition of a vaping prevention program in CHAMPS for 5th and 6th grade students is beneficial to adolescents, enhanced student learning regarding the dangers of vaping, and is a sustainable cur-

riculum. The goal was to provide early, educational intervention regarding vaping and the dangers of vaping to adolescents with the hope of deterring adolescents from beginning to vape or to stop vaping. From the feedback received from surveys, students were not fully aware of the dangers of vaping, the amount of nicotine in vaping, and harmful chemicals in the vape juice. Thus, the objective of educating adolescents regarding the dangers of vaping was achieved.

Positive consequences of this project include the transfer of knowledge from the adolescent to parents, with some parents contacting CHAMPS officers and the director for more information regarding vaping. The youth were listening and learning about the dangers of vaping and sharing with parents who had more questions.

Implementation of the vaping curriculum will be sustainable for the CHAMPS program. Deputies who participated in the program believe vaping is a genuine concern in middle and high school students, and hope to eventually expand the educational content regarding vaping. Minor edits will be completed before the implementation of the CHAMPS program in the next school year, including more images and videos to show the dangers of vaping and bringing the content to grade-appropriate level. Lastly, the author plans to continue to be a resource for vaping information and education if there is a need.

E-cigarette use in adolescents can be addictive, is dangerous, and may further exacerbate other dangerous behaviors. School-based educational programs are beneficial and impactful in reducing the incidence of vap-

ing in adolescents. Through a multi-disciplinary collaboration with a state agency, this project shared resources and combined talents to attempt to reduce vaping in adolescents. This project has proven participants perceived the intervention to be beneficial, impactful, and sustainable in counties across Georgia. The anticipated impact of this early education is to decrease the use of e-cigarettes in adolescents.

Limitations

Limitations of the study include sample size and response to the survey. Although 64 participants agreed to complete the survey, only 54 participants answered the entire survey. Another limitation was related to the oversight of administration of the curriculum due to the project being administered within a state agency. This made it difficult to assess how the vaping curriculum and survey were disseminated, and if every deputy who taught the vaping curriculum participated in the survey.

Conclusion and Practice Implications

This project demonstrates that health care providers can partner and collaborate with community or state agencies to have an impact on population health or reach a target audience. Using a school-based educational program can address unhealthy and harmful behavior, reach a target audience, and be beneficial and sustainable. This type of project could be applicable to elementary, middle, and high school students, and covers a variety of health-related topics so school-based educational interventions can have an impact or positive outcome to improve population health. Collaborating with other disciplines can benefit everyone in the community. ■

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